Validity Analysis: CRISISBENCH

Target context: Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)

Overall risk: **HIGH**

Deployment Context

Use case: Use case: Reliability of disaster reporting in news at national/local level, finding alignment with effective and verified information (e.g. noisy vs real signals). The evaluation data could report or not disaster damage.

Target population: Policy makers and government agencies in Argentina studying trustiness and reliability of disaster communication across different channels are (e.g news, social media post, press). They would be based in Buenos Aires, speakers of Rioplatense Spanish, with an intermediate to advanced level of English, at least a master's degree, and be part of an upper-middle-class household.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	3	Moderate gaps	high
Input Content ⚠	1	Serious concern	high
Input Form ✓	3	Moderate gaps	high
Output Ontology ⚠	2	Significant gaps	high
Output Content ⚠	2	Significant gaps	high
Output Form ✓	3	Moderate gaps	high
Average	**2.3**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

CrisisBench provides a methodologically rigorous, reproducible foundation for crisis-tweet classification with a clean two-task ontology (binary informativeness + multi-class humanitarian type) that aligns at the top level with international ISCRAM/OCHA-derived frameworks Argentine emergency professionals reference. However, three structural misalignments substantially limit its validity for the Buenos Aires policy-maker deployment: (1) Input Content (HIGH priority) is severely misaligned — 94.46% English [Q46], zero Argentine or Rioplatense Spanish content, and a 2010–2017 temporal window that predates the dissolution of Télam (July 2024) and creation of AFE (2025); the only Spanish content in the benchmark is Chilean and Venezuelan, not Rioplatense [DATASET-D2, D3, D6, D7]. (2) Output Ontology (HIGH priority) lacks the source credibility / signal-reliability dimension that is the deployment's core requirement; the user acknowledges this as a downstream construction step but the benchmark provides no learnable credibility signal. (3) Output Content shows annotator-population mismatch and empirically observable annotation noise, with evidence of systematic Spanish-language disaster-content under-recognition [DATASET-D9, D8]. Input Form, Input Ontology, and Output Form present moderate but tractable gaps. The benchmark is best treated as a usable signal-triage foundation for the user's planned downstream credibility-aggregation pipeline, not as a complete evaluation framework for the Argentine deployment.

ID	Evidence
Q46	"Among different languages of informativeness tweets, English tweets appear to be highest in the distribution compared to any other language, which is 94.46% of 156,899." (p.5)
D2	RT @FlorrGo: #FuerzaChile ,un 2 de abril ... Que vengan los ingleses a ayudarlos ahora ... A la larga o a la corta ,todo vuelve — Chilean Spanish earthquake solidarity tweet — not Argentine/Rioplatense
D3	Que lindo saber que en mi ciudad bella #Iquique hay gente con un corazón enorme dispuestos a ayudar... Emocionada — Chilean Spanish (Iquique city) solidarity tweet — Chilean, not Argentine
D6	IMPRESIONANTE terremoto Chile camara de seguridad farmacia - NEW VIDEO E...: http://t.co/bkUElKfFUj vía @YouTube — Chilean Spanish earthquake tweet — Chilean Spanish, not Rioplatense
D7	RT @marieli_d: Nuevamente se incendia #Amuay http://t.co/AiWlhgy3 — Venezuelan Spanish tweet about refinery explosion
D8	RT @meteomostoles: El tifón #BOPHA,de cat.4 y 947hPa en su centro,va camino de Filipinas.Lleva vientos sostenidos entre 2010 y 249km/h h ... — Spanish tweet about Philippines typhoon — meteorological content labelled not_humanitarian
D9	Ojo con el exceso de ayuda por los efectos del #TerremotoenChile porque puede producir mucho más obstáculos de lo que se cree. — Chilean Spanish warning about over-donation — labelled not_informative despite substantive crisis-relevant content

Practical Guidance

What This Benchmark Measures

CrisisBench measures English-language Twitter content's informativeness (binary) and humanitarian type (multi-class) for a fixed set of 2010–2017 disaster events concentrated in North America and globally reported crises. For the Argentine deployment, it provides a usable first-pass signal-triage capability — separating clear noise from clear crisis content at high accuracy on English content (0.866/0.829 F1) — that can serve as Step 1 of the user's planned downstream credibility-scoring pipeline. The strongest dimensions for the deployment are Input Form (text-only modality match) and the foundational utility of the binary informativeness task; the top-level humanitarian categories are recognizable to Argentine policy professionals via international taxonomy alignment.

Construct Depth

The benchmark probes informativeness and humanitarian type at moderate depth for English North American Twitter content but provides no depth on (a) source credibility, the deployment's core construct, or (b) Spanish-language or Rioplatense register. Cross-dataset F1 degradation of 14.3% [Q87] and ~8% multilingual humanitarian F1 drop [Q99] indicate that even the depth it does provide does not generalize cleanly across event contexts or languages. The dataset analysis reveals annotation noise that further limits the trustworthiness of the F1 numbers as indicators of real-world classification capability, particularly for borderline content where credibility decisions are most consequential.

What Else You Need

The deployment requires three supplementation streams: (1) Argentine/Rioplatense Spanish disaster content for fine-tuning or evaluation — addressing the Input Content gap; possibilities include collecting and annotating Argentine Twitter/X content using the CrisisBench taxonomy, leveraging the cross-lingual approach in arXiv:2209.02139 [WEB-17], or extending TREC-IS-style protocols [WEB-15] to Argentine events. (2) A credibility-scoring extension — addressing the Output Ontology gap; TREC-IS priority labels (Low/Medium/High/Critical) [WEB-15] are the closest precedent and could anchor downstream construction combining classifier outputs with Argentine source metadata (institutional affiliation, post-Télam press source registry, Buenos Aires Province Resolución 4/2025-compliant metadata). (3) Argentine-annotator label validation on a stratified sample — addressing the Output Content gap; required to quantify expected label disagreement, particularly for Spanish-language content where dataset analysis suggests systematic under-recognition risk.

ID	Evidence
Q87	“The cross dataset (i.e., train on CrisisLex and evaluate on CrisisNLP) results shows that there is a drop in performance. For example, there is 14.3% difference in F1 on CrisisNLP data using the CrisisLex model for the informativeness task.” (p.8)
Q99	“We observe that performance dropped significantly for the humanitarian task compared to English-only dataset. For example, ~8% drop using BERT model.” (p.9)
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)
WEB-17	Sánchez et al. 2022 cross-lingual crisis classification paper does not address Rioplatense register (https://arxiv.org/abs/2209.02139)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The top-level taxonomy (informativeness binary; humanitarian multi-class with categories like affected_individual, infrastructure_and_utilities_damage, requests_or_needs, response_efforts, caution_and_advice) is broadly compatible with the international ISCRAM/OCHA-derived frameworks Argentine emergency professionals use as reference points, and elicitation Q1 confirms users find these categories 'broadly sufficient'. However, the benchmark explicitly defers event-type classification [Q22, Q23], does not include sub-labels for Argentine hazard types (pampero, sudestada, ENSO drought, AMBA urban flash flood) or informal-settlement (villa) impact framing, and the absence of a Latin American taxonomy adaptation is confirmed at the field level [WEB-15]. Some humanitarian classes have very low support (e.g., 17 'terrorism related' tweets [Q52]), and categorical granularity varies significantly across constituent datasets [Q76]. Dataset analysis confirms the schema covers the core triage categories Argentine policy makers need [DATASET-D19, D20, D35, D42] but exposes the 'other_relevant_information' class as a heterogeneous catch-all that limits its utility for borderline credibility cases.

ID	Evidence
Q1	“We consolidate eight human-annotated datasets and provide 166.1k and 141.5k tweets for informativeness and humanitarian classification tasks, respectively.” (p.1)
Q22	“We only focused on two tasks for this study and we aim to address event types task in a future study.” (p.2)
Q23	“We consolidate eight datasets that were labeled for different disaster response classification tasks and whose labels can be mapped consistently for two tasks: informativeness and humanitarian information type classification.” (p.3)
Q52	“Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.” (p.5)
Q76	“The motivation of these experiments is to investigate whether model trained with consolidated dataset generalizes well across different sets. For the individual dataset classification experiments, we selected CrisisLex and CrisisNLP as they are relatively larger in size and have a reasonable number of class labels, i.e., six and eleven class labels, respectively.” (p.7)
D19	"Floods kill 3, 75,000 forced from Calgary homes http://t.co/c6OPFjFLkA — AP #news"
D20	As many as 100,000 people could be forced from their homes by heavy flooding in western Canada. Water levels peak today at noon. — Factual caution about Canadian floods
D35	I'M ASKING HELP FOR MY FAMILY WHICH IS VICTIM AND WHICH TAKE REFUGE IN PROVINCE. — Direct help request from disaster victim
D42	#Tacloban has 49 evacuation centers now sheltering over 30,000 people + @care is helping respond to #typhoonhagupit http://t.co/ePE9MsdpTh — Clear response effort tweet — CARE NGO involvement
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)

Strengths

- Top-level humanitarian categories (affected_individual, infrastructure_and_utilities_damage, requests_or_needs, response_efforts, caution_and_advice, donation_and_volunteering) align with ISCRAM-derived frameworks recognized by Argentine emergency management professionals, supporting first-pass triage utility
- Binary informativeness layer is structurally compatible with the deployment's first-pass signal-triage step and operationalizes a foundation for downstream credibility scoring

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required Argentine categories include pampero windstorm, sudestada/Río de la Plata flooding, ENSO-driven drought, urban AMBA flash flood, and villa (informal-settlement) impact framing. Top-level humanitarian categories (infrastructure damage, affected individuals, response efforts) are required and present.

IO-2: Argentine hazard sub-labels and informal-settlement impact framing are omitted. No Latin American adaptation of ISCRAM/OCHA taxonomies addressing pampero, sudestada, or ENSO-drought sub-types exists in the literature [WEB-15]. Event-type classification is deferred entirely [Q22, Q23].

IO-3: Some categories have negligible support (e.g., 'terrorism related' with only 17 tweets [Q52]; 'disease related' only in CrisisNLP [Q53]). The 'other_relevant_information' class is semantically incoherent per dataset analysis (Concern 7), reducing its utility.

IO-4: Documented gaps: (a) no event-type classification implemented; (b) no Argentine hazard sub-types; (c) no informal-settlement impact framing; (d) heterogeneous 'other_relevant_information' catch-all. These constitute construct underrepresentation for the deployment but the user has flagged them as desirable rather than blocking.

ID	Evidence
Q3	“Typical classification tasks in the community include (i) informativeness (i.e., informative vs. not-informative messages), (ii) humanitarian information types (e.g., affected individual reports, infrastructure damage reports), and (iii) event types (e.g., flood, earthquake, fire).” (p.1)
Q22	“We only focused on two tasks for this study and we aim to address event types task in a future study.” (p.2)
Q23	“We consolidate eight datasets that were labeled for different disaster response classification tasks and whose labels can be mapped consistently for two tasks: informativeness and humanitarian information type classification.” (p.3)
Q52	“Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.” (p.5)
Q53	“Similarly, the class "disease related" only appears in CrisisNLP.” (p.5)
Q76	“The motivation of these experiments is to investigate whether model trained with consolidated dataset generalizes well across different sets. For the individual dataset classification experiments, we selected CrisisLex and CrisisNLP as they are relatively larger in size and have a reasonable number of class labels, i.e., six and eleven class labels, respectively.” (p.7)
Q89	“Note that humanitarian task is a multi-class classification problem, which makes it a much more difficult task than the binary informativeness classification.” (p.8)
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)

Information Gaps

- Whether downstream sub-label extension to Argentine hazard types is feasible given absent training data

Requires Expert Verification

- Whether Argentine emergency management professionals would prioritize villa-impact sub-labels over generic affected_individual labels in operational triage

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Remediation

Gap 1: No sub-labels for Argentine hazard types (pampero, sudestada, ENSO drought, AMBA urban flash flood) or villa-impact framing

Recommendation: Extend the humanitarian taxonomy with Argentine hazard sub-labels and informal-settlement (villa) impact framing, validated with AMBA emergency management practitioners. Treat as a regional ontology overlay on top of CrisisBench's top-level categories rather than replacing them.

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content is the highest-priority dimension (HIGH per elicitation) and exhibits the most severe alignment failure. English tweets constitute 94.46% of the informativeness set [Q46], with non-English content largely confined to CrisisLex [Q47], and all classification experiments restricted to English [Q59]. The geographic distribution is concentrated in North American and globally reported events: Joplin tornado, Hurricane Sandy [Q27, Q28], 2010/2012 events [Q29], 2017 international disasters [Q31] — with no Argentine or Latin American disaster events documented. Dataset analysis empirically confirms that across 271 sampled examples, zero contain Argentine events, Argentine institutional actors, or Rioplatense Spanish; the only Spanish-language content is Chilean and Venezuelan [DATASET-D2, D3, D4, D6, D7]. The deployment's primary channel — Argentine Twitter/X with Rioplatense voseo, lunfardo, political satire, and references to local actors (SINAGIR, AFE, Buenos Aires province) — is entirely absent from the training distribution. The 2010–2017 temporal window [Q100] predates the dissolution of Télam (July 2024) [WEB-9], creation of AFE (2025) [WEB-10], and post-Musk Twitter/X conventions, meaning the institutional credibility landscape encoded in the data does not match the current Argentine deployment context.

ID	Evidence
Q27	"SWDM2013 dataset consists of data from two events: (i) the Joplin collection contains tweets from the tornado that struck Joplin, Missouri on May 22, 2011; (ii) The Sandy collection contains tweets collected from Hurricane Sandy that hit Northeastern US on Oct 29, 2012 (Imran et al. 2013a)." (p.3)
Q28	"ISCRAM2013 dataset consists of tweets from two different events occurred in 2011 (Joplin 2011) and 2012 (Sandy 2012). Note that this set of tweets are different than SWDM2013 set even though they are collected from same events (Imran et al. 2013b)." (p.3)
Q29	"Disaster Response Data (DRD) consists of tweets collected during various crisis events that took place in 2010 and 2012. This dataset is annotated using 36 classes that include informativeness as well as humanitarian categories." (p.3)
Q31	"CrisisMMD is a multimodal dataset consisting of tweets and associated images collected during seven disaster events that happened in 2017 (Alam et al. 2018). The annotations for this dataset is targeted for three classification tasks: (i) informative vs. not-informative, (ii) humanitarian categories (eight classes) and (iii) damage severity assessment." (p.3)
Q46	"Among different languages of informativeness tweets, English tweets appear to be highest in the distribution compared to any other language, which is 94.46% of 156,899." (p.5)
Q47	"Note that most of the non-English tweets appear in the CrisisLex dataset." (p.5)
Q59	"Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments." (p.6)
Q100	"The resulting dataset covers a time-span starting from 2010 to 2017, which can be used to study temporal aspects in crisis scenarios." (p.9)
D2	RT @FlorrGo: #FuerzaChile ,un 2 de abril ... Que vengan los ingleses a ayudarlos ahora ... A la larga o a la corta ,todo vuelve — Chilean Spanish earthquake solidarity tweet — not Argentine/Rioplatense
D3	Que lindo saber que en mi ciudad bella #Iquique hay gente con un corazón enorme dispuestos a ayudar... Emocionada — Chilean Spanish (Iquique city) solidarity tweet — Chilean, not Argentine
D4	#FuerzaChile jaja y piden ayuda jaja que los ayuden los ingleses manga de traidores — Chilean Spanish tweet expressing contempt for requesting outside help — not Argentine
D6	IMPRESIONANTE terremoto Chile camara de seguridad farmacia - NEW VIDEO E...: http://t.co/bkUElKfFUj vía @YouTube — Chilean Spanish earthquake tweet — Chilean Spanish, not Rioplatense
D7	RT @marieli_d: Nuevamente se incendia #Amuay http://t.co/AiWlhgy3 — Venezuelan Spanish tweet about refinery explosion
WEB-9	Télam dissolved by Decree 548/2024 in July 2024 — material change to Argentine formal press source landscape post-benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/309815/20240701)

WEB-10	AFE created by Decree 225/2025 — new federal emergency coordinator absent from benchmark institutional context (https://www.mhewc.org/argentina/)

Strengths

- Multi-event, multi-geography training distribution (20+ events spanning floods, earthquakes, hurricanes, typhoons, disease outbreaks, industrial accidents) provides exposure to diverse disaster registers, reducing categorical failure on hazard types absent from any single source [DATASET-D25, D27, D44]
- Some non-English Romance and Asian-language content (Italian, French, Portuguese, Tagalog, Spanish) is present and annotated, demonstrating the label schema has been exposed to multilingual data even though classification experiments are English-only [Q42, Q43, DATASET-D21, D39]

ID	Evidence
Q42	"Some of the existing datasets contain tweets in various languages (i.e., Spanish and Italian) without explicit assignment of a language tag." (p.5)
Q43	"In addition, many tweets have codeswitched (i.e., multilingual) content." (p.5)
D21	RT @frankowolf1: il comune di Mirandola cerca ing. e arch. contattare Polizia Municipale: 0535/611039, 800/197197 #terremoto — Italian tweet seeking engineers after earthquake — non-English caution
D25	After deadly Brazil nightclub fire, safety questions emerge. http://t.co/ah4JK2v2 — Brazil nightclub fire — closest geographic proximity to Argentina
D27	Torrential rains have damaged Kenya's infrastructure, severing road links between Nairobi and Garissa, where the air base of the World Food Programme (WFP) is located. — Africa infrastructure damage — internationally framed
D39	@lucilledizon: 517 ilang ilang st. bayanihan vilage cainta rizal, David Bautista +63 905 826 5094. Needs rescue @MMDA @govph #rescuePH — Tagalog rescue request — non-English, Philippine event, clear label
D44	Colorado floods shut down hundreds of oil and gas wells; recovery will take time: Colorado's oil and... http://t.co/HivwsVtrc8 #Oil #BRK — North American oil/gas infrastructure damage

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Argentine deployment requires Rioplatense Spanish lexical and pragmatic competence (voseo, lunfardo, political satire, local institutional actors). The benchmark contains no Argentine content [DATASET CRITICAL Concern 1]; the only Spanish content is Chilean/Venezuelan [DATASET-D2, D3, D4, D6, D7].

IC-2: Cultural alignment is poor for social media (the highest-priority channel per elicitation Q2). Argentine political satire of government crisis response, lunfardo register, and references to AMBA institutional actors are absent. Formal press content may align better but is also unrepresented, and the dissolution of Télam materially changes the post-2024 source landscape [WEB-8, WEB-9].

IC-3: Yes — the benchmark embeds North American institutional and cultural framing (US sports commentary, US political discourse, Sandy/Joplin/Maria contexts) [DATASET-D17, D18, D46]. The vernacular register learned as 'noise' is specifically North American social noise, not Argentine.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotators were involved in constructing CrisisBench. Recruiting Argentine annotators to flag culturally sensitive instances would require a separate validation effort.

IC-5: Documented content issues: (a) zero Argentine/Rioplatense content; (b) only Chilean/Venezuelan Spanish at small scale [WEB-16, WEB-17]; (c) 2010–2017 temporal window predates current Argentine institutional landscape [Q100, WEB-9, WEB-10]; (d) North American event and institutional dominance. These violate content validity for the deployment's primary social media channel.

ID	Evidence
Q2	“We provide benchmarks for both binary and multiclass classification tasks using several deep learning architectures including, CNN, fastText, and transformers.” (p.1)
Q27	“SWDM2013 dataset consists of data from two events: (i) the Joplin collection contains tweets from the tornado that struck Joplin, Missouri on May 22, 2011; (ii) The Sandy collection contains tweets collected from Hurricane Sandy that hit Northeastern US on Oct 29, 2012 (Imran et al. 2013a).” (p.3)
Q31	“CrisisMMD is a multimodal dataset consisting of tweets and associated images collected during seven disaster events that happened in 2017 (Alam et al. 2018). The annotations for this dataset is targeted for three classification tasks: (i) informative vs. not-informative, (ii) humanitarian categories (eight classes) and (iii) damage severity assessment.” (p.3)
Q46	“Among different languages of informativeness tweets, English tweets appear to be highest in the distribution compared to any other language, which is 94.46% of 156,899.” (p.5)
Q47	“Note that most of the non-English tweets appear in the CrisisLex dataset.” (p.5)
Q59	“Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments.” (p.6)
Q100	“The resulting dataset covers a time-span starting from 2010 to 2017, which can be used to study temporal aspects in crisis scenarios.” (p.9)
D2	RT @FlorrGo: #FuerzaChile ,un 2 de abril ... Que vengan los ingleses a ayudarlos ahora ... A la larga o a la corta ,todo vuelve — Chilean Spanish earthquake solidarity tweet — not Argentine/Rioplatense
D3	Que lindo saber que en mi ciudad bella #Iquique hay gente con un corazón enorme dispuestos a ayudar... Emocionada — Chilean Spanish (Iquique city) solidarity tweet — Chilean, not Argentine
D4	#FuerzaChile jaja y piden ayuda jaja que los ayuden los ingleses manga de traidores — Chilean Spanish tweet expressing contempt for requesting outside help — not Argentine
D6	IMPRESIONANTE terremoto Chile camara de seguridad farmacia - NEW VIDEO E...: http://t.co/bkUElKfFUj vía @YouTube — Chilean Spanish earthquake tweet — Chilean Spanish, not Rioplatense
D7	RT @marieli_d: Nuevamente se incendia #Amuay http://t.co/AiWlhgy3 — Venezuelan Spanish tweet about refinery explosion
D17	Tornado watch until 3am. Of course we just got our new garage door today. *sigh* — Personal complaint about tornado watch — Joplin 2011, North American event
D18	RT @tarnsnastylad: They need to name hurricanes better!.. superstorm Sandy? Seriously??.. Sounds like a FUCKING gay wrestler! — Crude personal opinion during Sandy — North American vernacular
D46	RT @eclecticbrotha: @AlGiordano Damn, I forgot how much of a shitshow Bernie's Puerto Rico operation was. — Political commentary during Hurricane Maria
WEB-8	https://en.wikipedia.org/wiki/T%C3%A9lam
WEB-9	Télam dissolved by Decree 548/2024 in July 2024 — material change to Argentine formal press source landscape post-benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/309815/20240701)
WEB-10	AFE created by Decree 225/2025 — new federal emergency coordinator absent from benchmark institutional context (https://www.mhewc.org/argentina/)
WEB-16	CrisisLex T6 used crowdsourced workers for relatedness labels; no Argentine annotators documented (https://crisislex.org/data-collections.html)
WEB-17	Sánchez et al. 2022 cross-lingual crisis classification paper does not address Rioplatense register (https://arxiv.org/abs/2209.02139)

Information Gaps

- Whether classifiers pre-trained on this benchmark transfer at any usable F1 level to Rioplatense Spanish content
- Whether the small Chilean subset would provide any positive transfer signal for Argentine deployment

Requires Expert Verification

- Argentine linguist/policy expert validation of which Rioplatense register features (voseo, lunfardo, political satire patterns) most threaten classification accuracy
- Operational assessment by AMBA emergency management staff of how WhatsApp and Telegram channel content (not in benchmark) interacts with the Twitter/X channel monitoring

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Remediation

Gap 1: Zero Argentine/Rioplatense Spanish content; 2010–2017 temporal window predates current Argentine institutional landscape (post-Télam, post-AFE)

Recommendation: Collect and annotate a stratified Argentine Twitter/X sample (covering pampero, sudestada, AMBA flash flood, ENSO drought, and villa-impact events) using the CrisisBench label schema; use this for fine-tuning and as a regional held-out evaluation set. Apply TREC-IS-style protocols [WEB-15] and the cross-lingual transfer approach from arXiv:2209.02139 [WEB-17].

ID	Evidence
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)
WEB-17	Sánchez et al. 2022 cross-lingual crisis classification paper does not address Rioplatense register (https://arxiv.org/abs/2209.02139)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is rated LOWER priority by elicitation, and the structural alignment is moderate. The benchmark is text-only Twitter content [Q38] matching the deployment's text-only modality, and Argentine professional cohorts have high digital infrastructure access [WEB-3, WEB-5]. Tokenization, URL/punctuation/user-id removal [Q38], and near-duplicate filtering at 0.75 cosine similarity [Q39, Q40] are reasonable preprocessing choices. However, three concerns remain: (a) the deployment spans Twitter/X, news articles, and press releases, but the benchmark is exclusively tweet-derived with no cross-channel transfer evaluation [WEB-20]; (b) language tagging is done via Google Cloud API [Q44, Q45] but lang_conf is missing for many CrisisMMD/AIDR examples (Concern 9), weakening language-stratified routing for the bilingual deployment; (c) all classification experiments are English-only [Q59] despite multilingual content being present in the consolidated data [Q42, Q43], so the form-level multilingual capability is not validated. The 70/10/20 split [Q60] and aggressive duplicate filtering [Q41, Q97] are methodological strengths.

ID	Evidence
Q38	"We first tokenize the text before applying any filtering. For tokenization, we used a modified version of the Tweet NLP tokenizer (O'Connor, Krieger, and Ahn 2010). Our modification includes lowercasing the text and removing URL, punctuation, and user id mentioned in the text. We then filter tweets having only one token." (p.4)
Q39	"We then use a similarity-based approach to remove the near-duplicates. To do this, we first convert the tweets into vectors using bag-of-ngram approach as a vector representation. We use uni- and bi-grams with their frequency-based representations. We then use cosine similarity to compute a similarity score between two tweets and flag them as duplicate if their similarity score is greater than the threshold value of 0.75." (p.4)
Q40	"To determine a plausible threshold value, we manually checked the tweets in different threshold bins (i.e., 0.70 to 1.0 with 0.05 interval) as shown in Figure 1, which we selected from consolidated informativeness dataset. By investigating the distribution and manual checking, we concluded that a threshold value of 0.75 is a reasonable choice." (p.4)
Q41	"As indicated in Table 1, there is a drop after filtering, e.g., ~25% for informativeness and ~20% for humanitarian tasks. It is important to note that failing to eradicate duplicates from the consolidated dataset would potentially lead to" (p.4)
Q42	"Some of the existing datasets contain tweets in various languages (i.e., Spanish and Italian) without explicit assignment of a language tag." (p.5)
Q43	"In addition, many tweets have codeswitched (i.e., multilingual) content." (p.5)
Q44	"We decided to provide a language tag for each tweet if it is not available with the respective dataset." (p.5)
Q45	"We used the language detection API of Google Cloud Services for this purpose." (p.5)
Q59	"Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments." (p.6)
Q60	"We split data into train, dev, and test sets with a proportion of 70%, 10%, and 20%, respectively, also reported in Table 5." (p.6)
Q97	"It is important to emphasize the fact that the results reported in this study are reliable as they are obtained on a dataset that has been cleaned from duplicate content, which might have led to misleading performance results otherwise." (p.9)
WEB-3	88.4% Argentine national internet penetration as of January 2024 (https://datareportal.com/reports/digital-2024-argentina)
WEB-5	AMBA professional cohort fully connected; 71% of fixed-line subscriptions in CABA/Buenos Aires province/Córdoba/Santa Fe/Mendoza (https://freedomhouse.org/country/argentina/freedom-net/2024)
WEB-20	TREC CrisisFACTS extends to multi-stream (Twitter, Reddit, Facebook, online news) but provides no tweet-to-news transfer benchmarks (https://crisisfacts.github.io/)

Strengths

- Text-only Twitter modality matches the deployment's text-only requirement; no signal-distribution mismatch at the modality level
- Rigorous near-duplicate filtering (0.75 cosine threshold, manually validated) reduces overestimated test results [Q39, Q40, Q97]
- Language tags assigned to all tweets via Google Cloud API enable downstream language-stratified analysis [Q44, Q45]

ID	Evidence
Q39	"We then use a similarity-based approach to remove the near-duplicates. To do this, we first convert the tweets into vectors using bag-of-ngram approach as a vector representation. We use uni- and bi-grams with their frequency-based representations. We then use cosine similarity to compute a similarity score between two tweets and flag them as duplicate if their similarity score is greater than the threshold value of 0.75." (p.4)
Q40	"To determine a plausible threshold value, we manually checked the tweets in different threshold bins (i.e., 0.70 to 1.0 with 0.05 interval) as shown in Figure 1, which we selected from consolidated informativeness dataset. By investigating the distribution and manual checking, we concluded that a threshold value of 0.75 is a reasonable choice." (p.4)
Q44	"We decided to provide a language tag for each tweet if it is not available with the respective dataset." (p.5)
Q45	"We used the language detection API of Google Cloud Services for this purpose." (p.5)
Q97	"It is important to emphasize the fact that the results reported in this study are reliable as they are obtained on a dataset that has been cleaned from duplicate content, which might have led to misleading performance results otherwise." (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Twitter text signal distributions match between benchmark and Argentine Twitter/X deployment at the form level. However, Twitter/X platform conventions have evolved post-2022 (post-Musk acquisition); pre-2017 tweet structure may differ in hashtag conventions, link formats, and length. INSUFFICIENT DOCUMENTATION on whether benchmark tokenization handles current X.com URL patterns.

IF-2: Argentine infrastructure supports text capture: 88.4% national internet penetration [WEB-3]; AMBA professional cohort has full connectivity [WEB-5]. No infrastructure obstacle.

IF-3: Cross-channel form gap: benchmark is tweet-only [Q38]; deployment also covers news articles and press releases. No cross-channel transfer benchmark exists in the literature [WEB-20]. Tweet-length constraints and Twitter-specific preprocessing (user-id removal, URL stripping) do not natively support longer-form institutional content.

IF-4: Documented form mismatches: (a) tweet-only vs. multi-channel deployment; (b) lang_conf missing for significant CrisisMMD/AIDR subset (Concern 9), reducing language-routing reliability; (c) temporal gap to current X.com platform conventions. These threaten external validity for non-Twitter deployment channels.

ID	Evidence
Q38	"We first tokenize the text before applying any filtering. For tokenization, we used a modified version of the Tweet NLP tokenizer (O'Connor, Krieger, and Ahn 2010). Our modification includes lowercasing the text and removing URL, punctuation, and user id mentioned in the text. We then filter tweets having only one token." (p.4)
Q42	"Some of the existing datasets contain tweets in various languages (i.e., Spanish and Italian) without explicit assignment of a language tag." (p.5)
Q59	"Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments." (p.6)

WEB-3	88.4% Argentine national internet penetration as of January 2024 (https://datareportal.com/reports/digital-2024-argentina)
WEB-5	AMBA professional cohort fully connected; 71% of fixed-line subscriptions in CABA/Buenos Aires province/Córdoba/Santa Fe/Mendoza (https://freedomhouse.org/country/argentina/freedom-net/2024)
WEB-20	TREC CrisisFACTS extends to multi-stream (Twitter, Reddit, Facebook, online news) but provides no tweet-to-news transfer benchmarks (https://crisisfacts.github.io/)

Information Gaps

- Whether benchmark-trained classifiers transfer to news article and press release form factors with any usable F1
- Whether lang_conf NA examples disproportionately misclassify by language

Requires Expert Verification

- Operational assessment of whether Argentine deployment teams can downstream-restrict input to tweets only or must accept news/press content

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Remediation

Gap 1: Tweet-only benchmark does not cover deployment's news article and press release channels; lang_conf missing for CrisisMMD/AIDR subsets weakens language routing

Recommendation: Conduct cross-channel transfer evaluation by testing CrisisBench-trained classifiers on a held-out Argentine news article and press release corpus. Backfill lang_conf scores for affected examples using a current language-identification model. Consider TREC CrisisFACTS [WEB-20] as a multi-stream framework reference.

ID	Evidence
WEB-20	TREC CrisisFACTS extends to multi-stream (Twitter, Reddit, Facebook, online news) but provides no tweet-to-news transfer benchmarks (https://crisisfacts.github.io/)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology is HIGH priority per elicitation. The two implemented tasks — binary informativeness and multi-class humanitarian type [Q12] — provide a usable foundation but do not natively include the source credibility, misinformation likelihood, or signal-to-noise reliability dimension that is the deployment's core requirement [elicitation Q3]. The user explicitly acknowledges this gap and intends to construct credibility scoring downstream by combining model outputs with external metadata, but the benchmark's data schema (id, event, source, text, lang, lang_conf, class_label) contains no tweet-level credibility metadata to learn from (CRITICAL Concern 2). Dataset analysis empirically confirms the disconnect: a 'DEATH TOLL Now under Investigation for COVER UP!!!' tweet labelled not_informative [DATASET-D45] and a Boston Bombing conspiracy tweet labelled informative [DATASET-D47] illustrate that informativeness ≠ credibility. The label space is also imbalanced even after consolidation (37.40% 'not humanitarian' [Q56]), with low-support classes [Q52, Q53, Q54] and class-mapping discrepancies that produce results reported only for classes the model can classify [Q88]. TREC-IS provides the closest precedent (Low/Medium/High/Critical priority labels) but covers no Argentine events [WEB-15].

ID	Evidence
Q3	“Typical classification tasks in the community include (i) informativeness (i.e., informative vs. not-informative messages), (ii) humanitarian information types (e.g., affected individual reports, infrastructure damage reports), and (iii) event types (e.g., flood, earthquake, fire).” (p.1)
Q12	“We provide benchmark results on English tweets set using state-of-the-art machine learning algorithms such as Convolutional Neural Networks (CNN), fastText (Joulin et al. 2017) and pre-trained transformer models (Devlin et al. 2019) for two classifications tasks, i.e., Informativeness (binary) and Humanitarian type (multi-class) classification.” (p.2)
Q52	“Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.” (p.5)
Q53	“Similarly, the class "disease related" only appears in CrisisNLP.” (p.5)
Q54	“The scarcity of the class labels poses a great challenge to design the classification model using individual datasets.” (p.5)
Q56	“For example, the distribution of "not humanitarian" is relatively higher (37.40%) than other class labels.” (p.5)
Q88	“In the humanitarian task, for different datasets in Table 6, we have different number of class labels. We report the results of those classes only for which the model is able to classify.” (p.8)
D45	https://t.co/najVJNt4Ip DEATH TOLL Now under Investigation for COVER UP!!! https://t.co/VuGGPYboIv — Credibility-ambiguous tweet about death toll cover-up labelled not_informative
D47	I was told I'm ignorant to think the Boston Bombing wasn't planned by the government to keep our attention away from schemes being planned. — Conspiracy tweet labelled informative
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)

Strengths

- Carefully documented label-mapping process across 8 source datasets [Q33, Q106–Q110, Q123–Q125] enables transparent inspection of the consolidated taxonomy
- Top-level humanitarian categories are oriented toward humanitarian aid relevance and align with internationally-normed disaster taxonomies Argentine professionals reference [Q18]

• Binary informativeness layer functions as a deployable first-pass triage filter — operationalizes Step 1 of the user's downstream credibility pipeline [DATASET-D34]

ID	Evidence
Q18	"In this study, we use the class labels that are important for humanitarian aid for disaster response tasks, which are common across the publicly available resources." (p.2)
Q33	"The datasets come with different class labels. We create a set of common class labels by manually mapping semantically similar labels into one cluster. For example, the label "building damaged," originally used in the AIDR system, is mapped to "infrastructure and utilities damage" in our final dataset." (p.3)
Q106	"In Table 12, we report class label mapping for ISCRAM2013, SWDM2013, CrisisLex and CrisisNLP datasets." (p.11)
Q107	"Note that all humanitarian class labels also mapped to informative, and not humanitarian labels mapped to not-informative in the data preparation step." (p.11)
Q108	"In Table 13, we report the class label mapping for informativeness and humanitarian tasks for DRD dataset." (p.11)
Q109	"The DSM dataset only contains tweets labeled as relevant vs not-relevant, which we mapped for informativeness task as shown in Table 14." (p.11)
Q110	"The CrisisMMD dataset has been annotated for informativeness and humanitarian task, therefore, very minor label mapping was needed as shown in Table in 15." (p.11)
Q123	"Table 12: Class label mapping and grouping for CrisisLex, CrisisNLP, ISCRAM2013, and SWDM2013 datasets. The symbol (✗) indicates we do not map the tweets with that label for this study." (p.13)
Q124	"The symbol (✗) indicates we do not map the tweets with that label for this study." (p.14)
Q125	"Table 16: Class label mapping for AIDR system." (p.15)
D34	RT @cityofcalgary: Acting Fire Chief, Ken Uzeloc, will address the media at Ogden Road and 17 Street S.E. at 10 p.m. #yyc #yycflood — Official municipal announcement — clear informative label, Canadian context

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Top-level humanitarian categories (affected_individual, infrastructure_damage, requests_or_needs, response_efforts, caution_and_advice, displaced_and_evacuations, donation_and_volunteering) are regionally relevant for Argentine emergency triage. The informativeness binary is also relevant.

OO-2: Critical missing category: source credibility / signal-reliability dimension absent from output space [DATASET CRITICAL Concern 2]. Argentine hazard sub-labels (pampero, sudestada) and villa-impact framing are absent. TREC-IS priority labels (Low/Medium/High/Critical) are the closest existing precedent but not part of CrisisBench [WEB-15].

OO-3: Some categories encode non-regional framing: 'terrorism related' (17 instances [Q52]) and 'disease related' (CrisisNLP only [Q53]) reflect specific source-event distributions. The 'not humanitarian' class at 37.40% [Q56] dominates and may inherit North American assumptions about what counts as humanitarian-relevant.

OO-4: Stakeholder-driven taxonomy redesign is warranted given the user-acknowledged credibility-scoring gap. The user's downstream construction approach (combining model outputs with external metadata) is a partial mitigation but not a benchmark-level fix.

OO-5: Documented taxonomy issues: (a) absent credibility/reliability dimension (structural validity violation for the deployment's core construct); (b) absent Argentine sub-labels; (c) heterogeneous 'other_relevant_information' catch-all (Concern 7); (d) imbalanced classes with very low minority support [Q52–Q56].

ID	Evidence
Q12	“We provide benchmark results on English tweets set using state-of-the-art machine learning algorithms such as Convolutional Neural Networks (CNN), fastText (Joulin et al. 2017) and pre-trained transformer models (Devlin et al. 2019) for two classifications tasks, i.e., Informativeness (binary) and Humanitarian type (multi-class) classification.” (p.2)
Q18	“In this study, we use the class labels that are important for humanitarian aid for disaster response tasks, which are common across the publicly available resources.” (p.2)
Q52	“Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.” (p.5)
Q53	“Similarly, the class "disease related" only appears in CrisisNLP.” (p.5)
Q54	“The scarcity of the class labels poses a great challenge to design the classification model using individual datasets.” (p.5)
Q55	“Even after combining the datasets, the imbalance in class distribution seems to persist (last column in Table 4).” (p.5)
Q56	“For example, the distribution of "not humanitarian" is relatively higher (37.40%) than other class labels.” (p.5)
Q88	“In the humanitarian task, for different datasets in Table 6, we have different number of class labels. We report the results of those classes only for which the model is able to classify.” (p.8)
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)
WEB-21	2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified (https://arxiv.org/html/2410.21360v2)

Information Gaps

- Whether downstream credibility-scoring pipelines built on top of this benchmark's outputs achieve operationally useful F1 in Argentine deployment
- Whether external metadata (institutional affiliation, source popularity) the user envisions combining with model outputs is reliably available in Argentine post-Télam press environment

Requires Expert Verification

- Argentine emergency management expert validation of which output categories are operationally critical vs. nice-to-have
- Whether TREC-IS priority labels would be a useful supplementation target for Argentine deployment

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Remediation

Gap 1: No source credibility / signal-reliability dimension in the output space, which is the deployment's core requirement

Recommendation: Construct a downstream credibility-scoring layer combining CrisisBench classifier outputs with Argentine source-metadata streams (post-Télam press source registry, institutional affiliation, AFE/SINAGIR official channels, account history). Use TREC-IS priority labels (Low/Medium/High/Critical) [WEB-15] as the target output schema; align with Buenos Aires Province Resolución 4/2025 [WEB-11] for regulatory compliance.

ID	Evidence
WEB-11	https://regulations.ai/regulations/argentina-summary
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation provenance is minimally documented: the only directly attested annotation protocol is that AIDR data was labeled by domain experts [Q111]. For the remaining seven source datasets, no annotator demographics, recruitment methods, or inter-annotator agreement statistics are reported in the paper. Web research confirms CrisisNLP used crowdsourced workers with three-annotator agreement [WEB-18], but no Argentine annotators or Argentine-norm calibration is documented anywhere in the provenance chain. Cross-dataset evaluation reveals a 14.3% F1 drop when training on CrisisLex and evaluating on CrisisNLP [Q87], demonstrating that label conventions are not fully portable across event contexts — which raises substantial concern about portability to Argentine discourse. Dataset analysis empirically reveals concrete annotation noise: a Chilean cinema tweet labelled displaced_and_evacuations [DATASET-D1], NGO mission descriptions labelled requests_or_needs [DATASET-D37, D38], substantive typhoon warnings labelled not_informative [DATASET-D28, D30], philosophical text labelled informative [DATASET-D11], and Spanish-language disaster warnings systematically labelled not_informative [DATASET-D9, D8] — suggesting potential systematic under-recognition of Spanish-language disaster signals. Elicitation Q4 indicates Argentine professionals largely accept Global North label conventions, which softens but does not eliminate the concern.

ID	Evidence
Q4	"First, few efforts have been invested to develop standard datasets (specifically, train/dev/test splits) and benchmarks for the community to compare their results, models, and techniques." (p.1)
Q87	"The cross dataset (i.e., train on CrisisLex and evaluate on CrisisNLP) results shows that there is a drop in performance. For example, there is 14.3% difference in F1 on CrisisNLP data using the CrisisLex model for the informativeness task." (p.8)
Q111	"The AIDR data has been labeled by domain experts using AIDR system and has been labeled during different events." (p.11)
D1	El trabajo de la mejor Daniela López que conozco en pantalla grande #mataraunhombe #chile #cinema… http://t.co/asAx6KqNvD — Spanish tweet about Chilean film, labelled displaced/evacuations — apparent annotation error
D8	RT @meteomostoles: El tifón #BOPHA,de cat.4 y 947hPa en su centro,va camino de Filipinas.Lleva vientos sostenidos entre 2010 y 249km/h h ... — Spanish tweet about Philippines typhoon — meteorological content labelled not_humanitarian
D9	Ojo con el exceso de ayuda por los efectos del #TerremotoenChile porque puede producir mucho más obstáculos de lo que se cree. — Chilean Spanish warning about over-donation — labelled not_informative despite substantive crisis-relevant content
D11	when the ambushes of this century, hold closes even when life puts you in discomfiture, hold closes even that one of condescends your family, so God will facilitate you shortcoming — Philosophical/religious text labelled informative — no disaster content
D28	A tropical cyclone will affect my area. Winds of greater than 100 kph up to 185 kph may be expected in at least 18 hours. #TyphoonHagupit — Substantive typhoon warning labelled not_informative — clear label inconsistency
D30	Bangladesh:palazzo non era per fabbriche: Il progetto del Rana Plaza prevedeva solo 6 piani non 9 o 10 http://t.co/VG9qCqa3xX — Italian investigative tweet about Rana Plaza building code — labelled not_informative
D37	Equipped with a power generator and air conditioner, the tents can also house an emergency medical center. — Capability description mislabelled as requests_or_needs — annotation inconsistency
D38	AmeriCares solicits donations of medicines, medical supplies, and other relief materials from manufacturers, and delivers them quickly and reliably to indigenous health and welfare professionals in 137 countries around the world. — NGO capability description labelled requests_or_needs — mislabel
WEB-18	CrisisNLP used crowdsourced annotation with three-annotator agreement; demographics not published (https://aclanthology.org/L16-1259.pdf)

Strengths

- Argentine policy professionals draw on international (Global North) disaster communication standards as reference points [elicitation Q4], reducing expected systematic label disagreement for formal content
- AIDR subset has documented domain-expert annotation [Q111], providing a higher-confidence subset for evaluation

ID	Evidence
Q4	“First, few efforts have been invested to develop standard datasets (specifically, train/dev/test splits) and benchmarks for the community to compare their results, models, and techniques.” (p.1)
Q111	“The AIDR data has been labeled by domain experts using AIDR system and has been labeled during different events.” (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially — international (OCHA/ISCRAM-derived) label conventions used in the benchmark are recognized by Argentine emergency professionals [elicitation Q4], but Argentine institutional and journalistic norms for what constitutes a 'real signal' of disaster impact are not directly represented. Edge cases in political framing and informal-channel content are not aligned.

OC-2: Likely disagreement at margins: Argentine political satire of government crisis response and lunfardo-inflected commentary may be misclassified. Dataset analysis shows Spanish-language disaster warnings labelled not_informative [DATASET-D9, D8], suggesting systematic Spanish-content under-recognition that would propagate to Argentine deployment.

OC-3: Annotator demographics are not reported in the paper for seven of eight source datasets. Only AIDR is documented as domain-expert annotated [Q111]. CrisisNLP used crowdsourcing with three-annotator agreement [WEB-18] but demographics are not published. No Argentine annotators are documented.

OC-4: Re-annotation by an Argentine annotator pool would be advisable for any deployment-critical evaluation; this is user-acknowledged as a gap (elicitation flag).

OC-5: INSUFFICIENT DOCUMENTATION — paper does not describe aggregation methods or how minority annotator perspectives were handled across the eight constituent datasets.

OC-6: Documented label issues: (a) annotator demographics undocumented for 7/8 sources; (b) 14.3% cross-dataset F1 drop signals non-portable conventions [Q87]; (c) empirically observable annotation noise [DATASET-D1, D37, D38, D28, D33]; (d) Spanish-content label asymmetry [DATASET-D9, D8]. These violate convergent and external validity for Argentine deployment.

ID	Evidence
Q4	“First, few efforts have been invested to develop standard datasets (specifically, train/dev/test splits) and benchmarks for the community to compare their results, models, and techniques.” (p.1)
Q87	“The cross dataset (i.e., train on CrisisLex and evaluate on CrisisNLP) results shows that there is a drop in performance. For example, there is 14.3% difference in F1 on CrisisNLP data using the CrisisLex model for the informativeness task.” (p.8)
Q94	“Social media data is noisy and it often poses a challenge for labeling and training classifiers.” (p.9)
Q111	“The AIDR data has been labeled by domain experts using AIDR system and has been labeled during different events.” (p.11)

D1	El trabajo de la mejor Daniela López que conozco en pantalla grande #mataraunhombe #chile #cinema… http://t.co/asAx6KqNvD — Spanish tweet about Chilean film, labelled displaced/evacuations — apparent annotation error
D8	RT @meteomostoles: El tifón #BOPHA,de cat.4 y 947hPa en su centro,va camino de Filipinas.Lleva vientos sostenidos entre 2010 y 249km/h h ... — Spanish tweet about Philippines typhoon — meteorological content labelled not_humanitarian
D9	Ojo con el exceso de ayuda por los efectos del #TerremotoenChile porque puede producir mucho más obstáculos de lo que se cree. — Chilean Spanish warning about over-donation — labelled not_informative despite substantive crisis-relevant content
D28	A tropical cyclone will affect my area. Winds of greater than 100 kph up to 185 kph may be expected in at least 18 hours. #TyphoonHagupit — Substantive typhoon warning labelled not_informative — clear label inconsistency
D33	Check out these award winning Long Island renovations: #longisland #suffolk https://t.co/wLnIbOENK9 — Completely unrelated home renovations tweet in other_relevant_information
D37	Equipped with a power generator and air conditioner, the tents can also house an emergency medical center. — Capability description mislabelled as requests_or_needs — annotation inconsistency
D38	AmeriCares solicits donations of medicines, medical supplies, and other relief materials from manufacturers, and delivers them quickly and reliably to indigenous health and welfare professionals in 137 countries around the world. — NGO capability description labelled requests_or_needs — mislabel
WEB-16	CrisisLex T6 used crowdsourced workers for relatedness labels; no Argentine annotators documented (https://crisislex.org/data-collections.html)
WEB-18	CrisisNLP used crowdsourced annotation with three-annotator agreement; demographics not published (https://aclanthology.org/L16-1259.pdf)

Information Gaps

- Annotator demographics, education levels, and recruitment for 7 of 8 source datasets
- Inter-annotator agreement statistics across source datasets
- Aggregation methods used to resolve disagreement across multi-annotator labels
- Whether the observed Spanish-language label asymmetry is systematic or sampling artifact

Requires Expert Verification

- Argentine policy/journalism expert re-annotation of a stratified Spanish-language sample to quantify expected label disagreement
- Validation of whether AIDR's domain-expert subset performs differently on Argentine-relevant content than crowdsourced subsets

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Remediation

Gap 1: Undocumented annotator demographics for 7 of 8 source datasets; observed annotation noise and systematic Spanish-language label asymmetry

Recommendation: Conduct re-annotation of a stratified Spanish-language and Argentine-political-satire sample by an Argentine policy/journalism expert pool to quantify expected label disagreement. Restrict deployment-critical evaluation to AIDR domain-expert-annotated subset where possible. Publish a Datasheet documenting Argentine annotator demographics and IAA.

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is MODERATE priority per elicitation. The benchmark produces discrete classification labels evaluated via weighted precision/recall/F1 [Q75], with documented model families (CNN [Q63], fastText [Q64], BERT/DistilBERT/RoBERTa [Q66]) achieving 0.866 F1 informativeness and 0.829 F1 humanitarian on consolidated English test set [Q84]. Hyperparameters are fully documented [Q113–Q121] and ten re-runs are performed for stability [Q72]. The output form is clean, reproducible, and suitable for downstream pipeline construction. However, two structural gaps remain: (a) the deployment envisions an aggregated continuous credibility score, which the benchmark does not produce — it requires downstream construction (user-acknowledged); no published pipeline combining CrisisBench outputs with external metadata exists [WEB-21]; (b) multilingual evaluation shows ~8% F1 drop on humanitarian classification using BERT [Q99], indicating that even where multilingual output is attempted, performance degrades substantially. Pre-trained models are mainly trained on non-Twitter text, an acknowledged limitation [Q65]. The discrete-to-continuous transformation is a structural gap but is bounded and tractable.

ID	Evidence
Q63	"The current state-of-the-art disaster classification model is based on the CNN architecture." (p.6)
Q64	"For the fastText (Joulin et al. 2017), we used pre-trained embeddings trained on Common Crawl, which is released by fastText for English." (p.6)
Q65	"Though the pre-trained models are mainly trained on non-Twitter text, we hypothesize that their rich contextualized embeddings would be beneficial for the disaster domain." (p.6)
Q66	"In this work, we choose the pre-trained models BERT (Devlin et al. 2019), DistilBERT (Sanh et al. 2019), and RoBERTa (Liu et al. 2019) for the classification tasks." (p.6)
Q72	"Due to the instability of the pre-trained models as reported in (Devlin et al. 2019), we perform ten re-runs for each experiment using different seeds, and we select the model that performs best on the dev set." (p.6)
Q75	"To measure the performance of each classifier, we use weighted average precision (P), recall (R), and F1-measure (F1). The rationale behind choosing the weighted metric is that it takes into account the class imbalance problem." (p.7)
Q84	"The model trained using the consolidated dataset achieves 0.866 (F1) for informativeness and 0.829 for humanitarian, which is better than the models trained using individual datasets." (p.8)
Q99	"We observe that performance dropped significantly for the humanitarian task compared to English-only dataset. For example, ~8% drop using BERT model." (p.9)
Q113	"In this section, we report parameters for CNN and BERT model." (p.11)
Q114	"All experimental scripts will be publicly Hyper-parameters include:" (p.11)
Q115	"Batch size: 8" (p.11)
Q116	"Number of epochs: 10" (p.11)
Q117	"Max seq length: 128" (p.11)
Q118	"Learning rate (Adam): 2e-5" (p.11)
Q119	"BERT (bert-base-uncased): L=12, H=768, A=12, total parameters = 110M; where L is number of layers (i.e., Transformer blocks), H is the hidden size, and A is the number of self-attention heads." (p.11)
Q120	"DistilBERT (distilbert-base-uncased): it is a distilled version of the BERT model consists of 6-layer, 768-hidden, 12-heads, 66M parameters." (p.11)

Q121	“RoBERTa (roberta-large): it is using the BERT-large architecture consists of 24-layer, 1024-hidden, 16-heads, 355M parameters.” (p.11)
WEB-21	2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified (https://arxiv.org/html/2410.21360v2)

Strengths

- Standard, reproducible evaluation methodology with weighted F1 chosen explicitly to handle class imbalance [Q75]
- Multiple model families benchmarked (CNN, fastText, transformers) with full hyperparameter documentation [Q113–Q121] and stability via 10 re-runs [Q72]
- Discrete classification labels are easily consumed by downstream credibility-aggregation pipelines, supporting the user's planned architecture

ID	Evidence
Q72	“Due to the instability of the pre-trained models as reported in (Devlin et al. 2019), we perform ten re-runs for each experiment using different seeds, and we select the model that performs best on the dev set.” (p.6)
Q75	“To measure the performance of each classifier, we use weighted average precision (P), recall (R), and F1-measure (F1). The rationale behind choosing the weighted metric is that it takes into account the class imbalance problem.” (p.7)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial mismatch: benchmark produces discrete classification labels; deployment envisions an aggregated continuous credibility score combining model outputs with external metadata. The user acknowledges this requires downstream construction. No published end-to-end credibility pipeline applicable to the deployment exists [WEB-21].

OF-2: Not applicable — deployment is text-only; no text-to-speech requirement.

OF-3: Target cohort is high-literacy (master's degree+) [WEB-1, WEB-2]; literacy is not a constraint. Output form does not need accessibility adaptations for the analyst cohort.

OF-4: Documented form mismatches: (a) discrete labels vs. envisioned continuous credibility score (user-acknowledged downstream construction step); (b) ~8% F1 drop in multilingual humanitarian classification [Q99] — output reliability degrades for non-English content. These threaten external validity for the operational credibility-scoring use case but are bounded.

ID	Evidence
Q65	“Though the pre-trained models are mainly trained on non-Twitter text, we hypothesize that their rich contextualized embeddings would be beneficial for the disaster domain.” (p.6)
Q84	“The model trained using the consolidated dataset achieves 0.866 (F1) for informativeness and 0.829 for humanitarian, which is better than the models trained using individual datasets.” (p.8)
Q99	“We observe that performance dropped significantly for the humanitarian task compared to English-only dataset. For example, ~8% drop using BERT model.” (p.9)
WEB-1	Argentine national literacy 98.08% (INDEC); target cohort effectively 100% (https://www.perfil.com/noticias/sociedad/el-censo-argentino-arrojo-98-de-alfabetizacion-en-el-pais-pero-no-se-refleja-en-las-aulas.phtml)
WEB-2	https://es.theglobaleconomy.com/Argentina/Literacy_rate/
WEB-21	2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified (https://arxiv.org/html/2410.21360v2)

Information Gaps

- Whether downstream aggregation of CrisisBench classifier outputs with external Argentine source-metadata produces operationally useful credibility scores
- Whether the ~8% multilingual F1 drop generalizes to or worsens for Rioplatense Spanish

Requires Expert Verification

- Validation by Argentine policy makers of what continuous credibility score thresholds are operationally meaningful for their decision-making

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Remediation

Gap 1: Discrete labels do not natively yield the aggregated continuous credibility score the deployment envisions; ~8% F1 drop in multilingual setting

Recommendation: Design the downstream credibility-aggregation function explicitly: define how classifier confidence scores combine with source-popularity, institutional-affiliation, and corroboration metadata to yield a continuous score. Validate score thresholds with AMBA policy makers and document the transformation pipeline for transparency under Argentine AI governance instruments [WEB-11, WEB-12].

ID	Evidence
WEB-11	https://regulations.ai/regulations/argentina-summary
WEB-12	https://digital.nemko.com/regulations/ai-regulation-argentina